

TASK 3 - REPORT

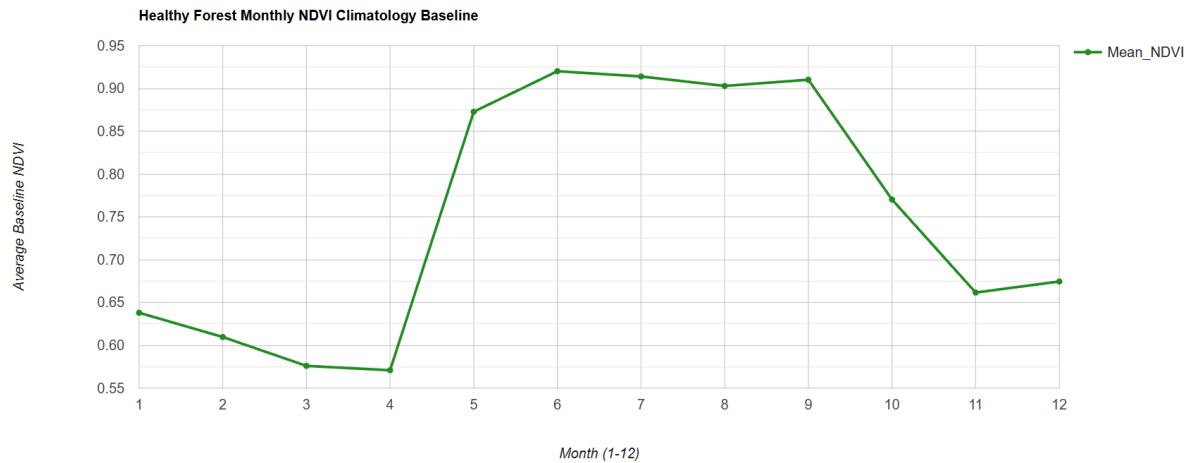
1. Monthly NDVI Climatology Baseline

Pixel used:

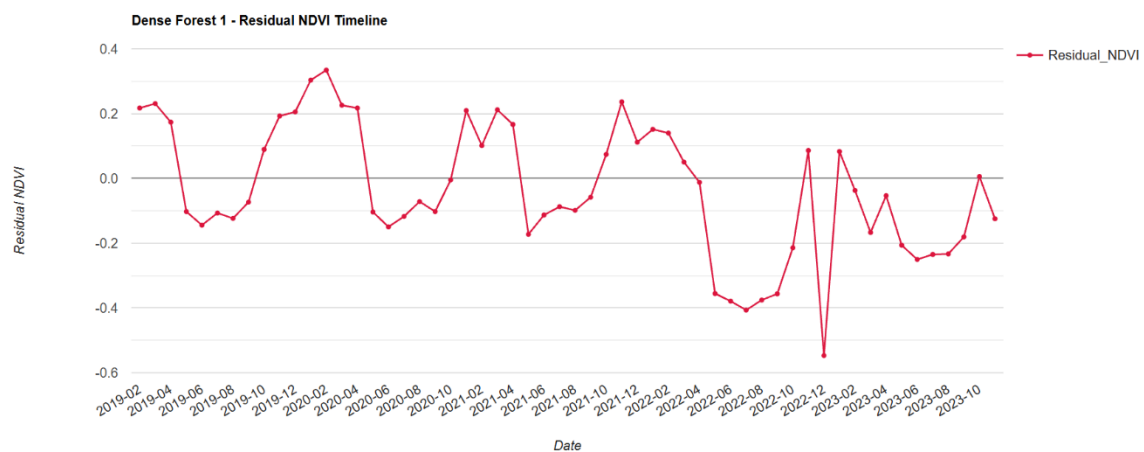
Healthy Forest - (11.73603, 47.9334)

Month	Baseline NDVI
January	0.638
February	0.610
March	0.576
April	0.571
May	0.873
June	0.920
July	0.914
August	0.903
September	0.910
October	0.770
November	0.662
December	0.674

Healthy Forest Monthly NDVI Climatology Baseline Graph:



A) Dense Forest Pixel (1) – [LOSS YEAR: 2022]



Dense Forest Pixel (1) Residual NDVI Timeline

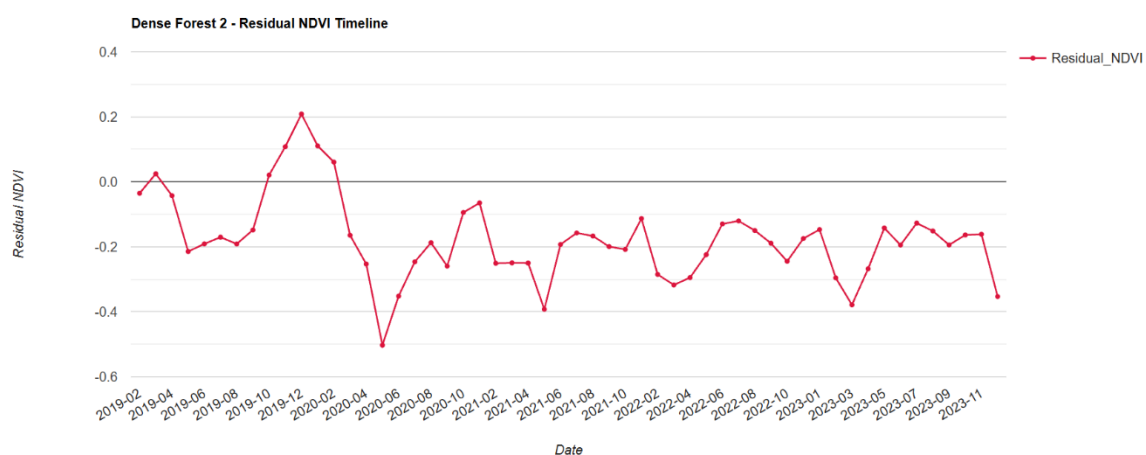
Observation:

A major change occurs during May 2022, where the residual drops sharply from values close to zero to approximately -0.36 to -0.41 throughout the entire 2022 growing season. The most extreme deviation occurs in December 2022 (-0.547). Although there is slight recovery

during 2023, the residuals remain consistently negative (roughly -0.18 to -0.25 during the growing season), indicating that the vegetation never returned to its expected healthy seasonal condition.

***Yes. The clearing event stands out much more clearly in the residual series than in the raw NDVI series.**

B) Dense Forest Pixel (2) – [LOSS YEAR: 2020]



Dense Forest Pixel (2) Residual NDVI Timeline

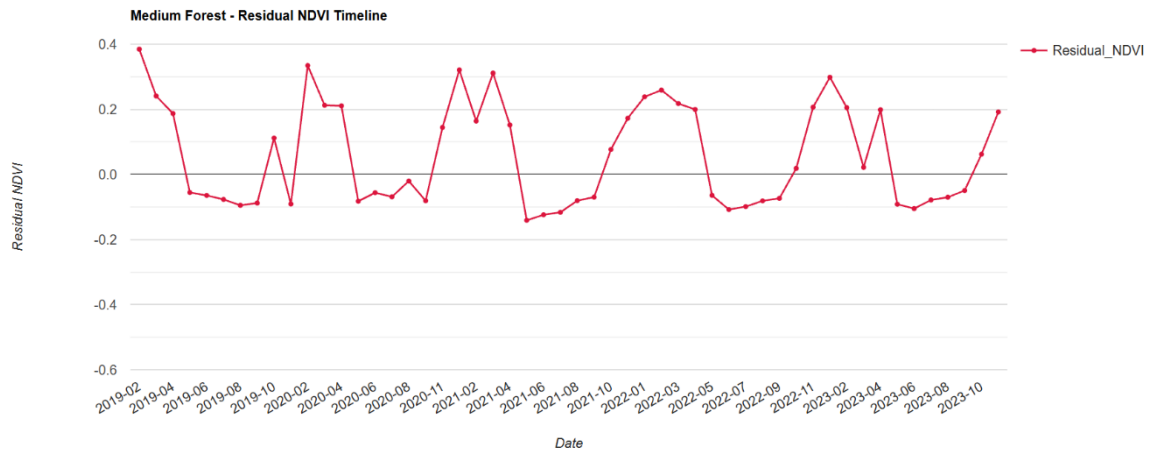
Observation:

Before the disturbance (2019), the residuals fluctuate around the healthy forest baseline, with only moderate negative deviations during the growing season (approximately -0.17 to -0.22) and positive deviations during winter. This indicates that although this forest naturally had slightly lower NDVI than the healthy reference, its seasonal behaviour remained relatively stable.

After the Hansen confirmed 2020 loss year, the residual series shows a significant negative anomaly beginning in March - April 2020 and becoming strongest during May 2020 (-0.503). Unlike normal seasonal fluctuations, these negative residuals persist throughout the remainder of 2020 and continue across 2021, 2022, and 2023, with most growing-season residuals remaining between -0.15 and -0.40. The forest never returns to its pre-disturbance baseline, indicating a long-term reduction in vegetation greenness.

*** Yes. The clearing event is more clearly distinguished in the residual series than in the raw NDVI series.**

C) Medium Forest Pixel – [LOSS YEAR: 2022]



Medium Forest Pixel Residual NDVI Timeline

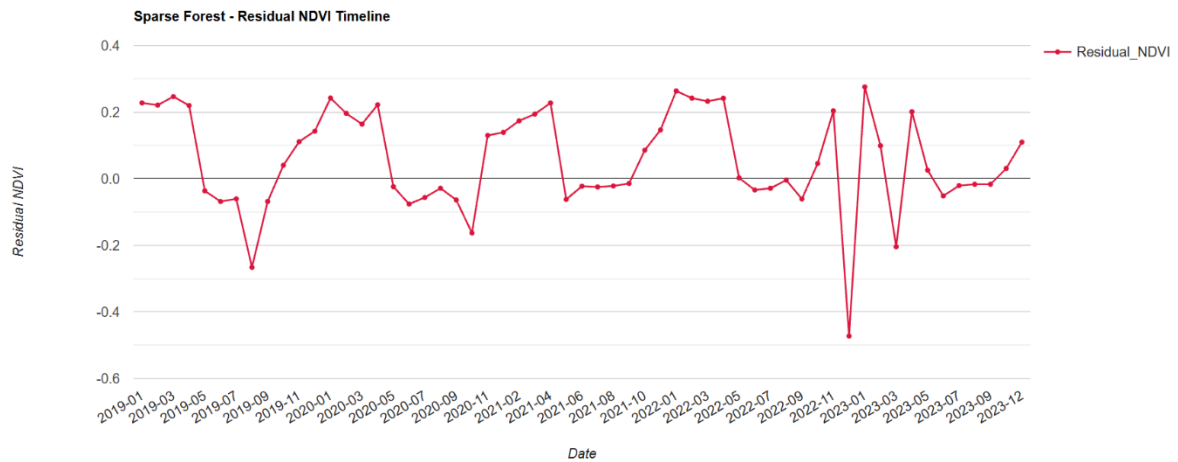
Observation:

The residual NDVI values remain relatively stable throughout the study period. Before the reported loss year (2019 - 2021), the pixel follows a consistent seasonal pattern, with positive residuals during winter months and only modest negative residuals during the growing season (typically -0.05 to -0.14).

Around the Hansen-confirmed 2022 loss year, the residual series does not exhibit a sudden or sustained negative anomaly. The growing-season residuals during 2022 remain similar to those observed before the reported disturbance (approximately -0.06 to -0.11), while winter residuals remain positive. During 2023, the pattern also remains comparable, with only a brief reduction in March before returning to values similar to previous years.

***No. The clearing event does not stand out more clearly in the residual series than in the raw NDVI series.**

D) Sparse Forest Pixel - [LOSS YEAR:2020]



Sparse Forest Pixel Residual NDVI Timeline

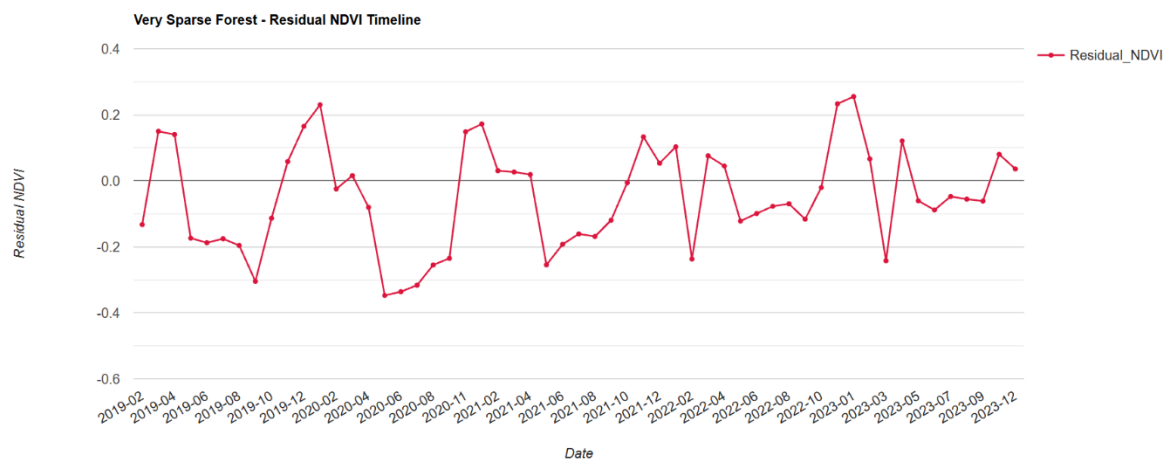
Observation:

The residual NDVI series shows no clear evidence of a sustained canopy disturbance during the Hansen-reported loss year (2020). Residual values remain close to the expected monthly climatology throughout 2020, indicating that the observed NDVI follows the normal seasonal behaviour of the healthy reference forest.

A pronounced negative residual occurs in December 2022 (-0.473), followed immediately by a strong recovery in January 2023 (+0.276). Because the anomaly lasts for only a single monthly observation and is followed by an immediate return toward baseline conditions, it is unlikely to represent permanent forest clearing.

***No. The clearing event does not stand out more clearly in the residual series than in the raw NDVI series.**

E) Very Sparse Forest Pixel - [LOSS YEAR: 2020]



Very Sparse Forest Pixel Residual NDVI Timeline

Observation:

Unlike the dense forest pixels, the residual NDVI series does not display a single sharp negative anomaly centered on the reported Hansen loss year (2020). Instead, the residuals are already variable before 2020 and remain variable afterwards.

Around the reported loss year, the residuals become moderately negative during the 2020 growing season (approximately -0.35 to -0.24 between May and September 2020), indicating vegetation conditions below the expected monthly baseline. However, this decline is not isolated because similar negative departures reappear during 2021 and again during parts of 2022 and 2023.

***No. The clearing event does not stand out more clearly in the residual series than in the raw NDVI series.**

Final verdict:

Monthly climatology residuals effectively remove seasonal phenology and improve the detection of persistent canopy loss in dense forests, but provide limited additional benefit for medium and sparse forests where natural variability and mixed land-cover effects dominate the NDVI signal.

GEE Code Link: <https://code.earthengine.google.com/337964255f057817d911a519c34dc3cf>

GitHub Repository Link: <https://github.com/SiddharthBhimpure/bavaria-forest-change-analysis.git>